

VLAN, DHCP & TRUNK NETWORK CONFIGURATION

Mini Case Study: Vlaminck Textiles

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Program: B.Tech CSE

Subject: Computer Networking & Cyber Security

Assignment: DHCP, VLANs and Trunk Links

Tool Used: Cisco Packet Tracer

1. AIM

To design and configure a network using VLANs, DHCP and an 802.1Q trunk link in Cisco Packet Tracer.

The network is divided into four VLANs:

- VLAN 10 – Sales
 - VLAN 20 – Finance
 - VLAN 30 – Manufacturing
 - VLAN 99 – Guest Wi-Fi
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2. OBJECTIVES

1. To create separate VLANs for different departments.
2. To configure DHCP for automatic IP address assignment.
3. To configure an 802.1Q trunk between SW1 and SW2.
4. To allow VLANs 10, 20, 30 and 99 over the trunk.
5. To configure VLAN 999 as the native VLAN.
6. To verify DHCP address allocation.
7. To test connectivity between each server and its corresponding router.

3. NETWORK TOPOLOGY

The network was implemented according to the topology explained by the faculty.

The setup consists of:

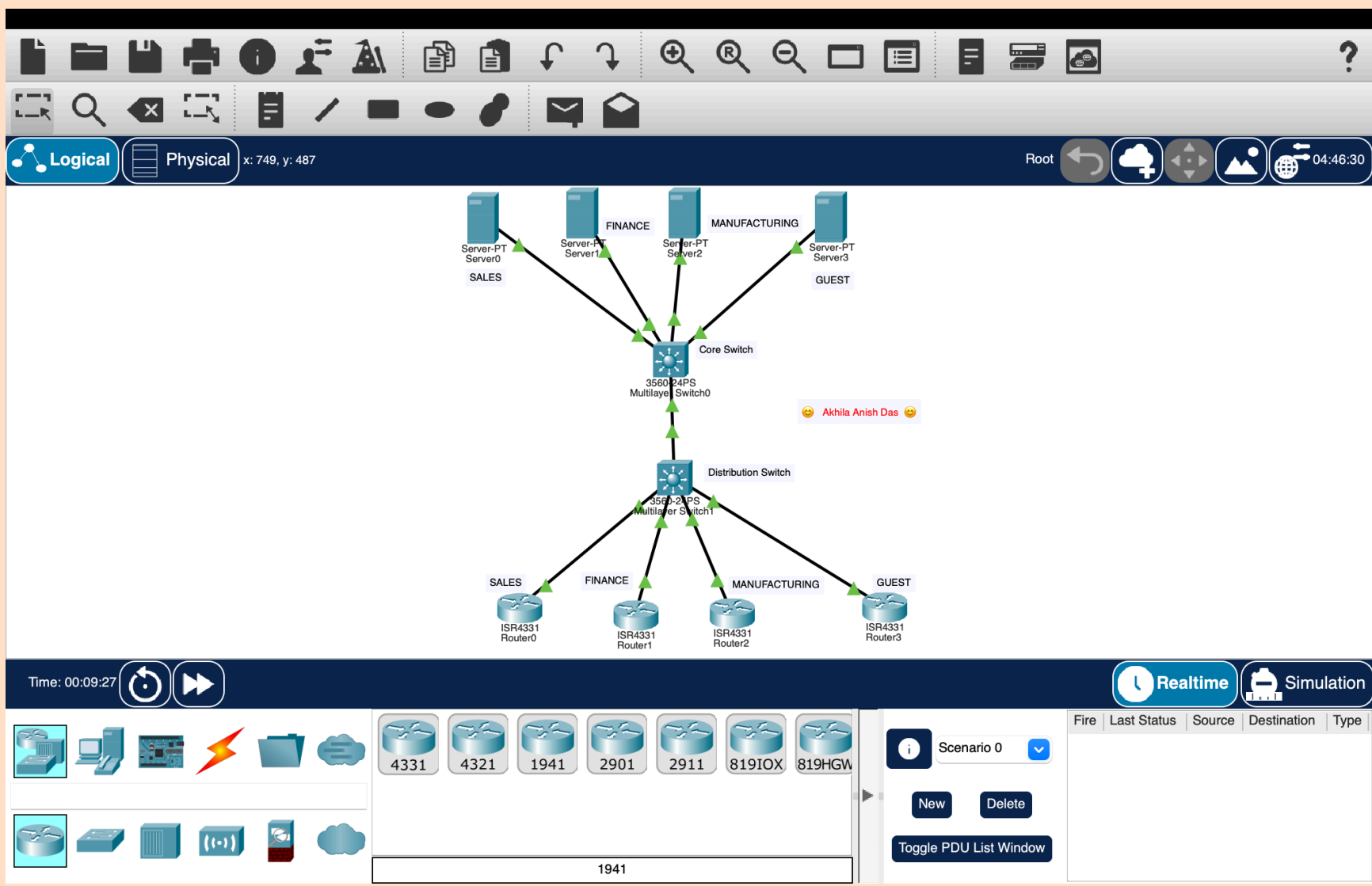
- 4 Servers: S1, S2, S3 and S4
- 2 Switches: SW1 and SW2
- 4 Routers: R1, R2, R3 and R4

Device and VLAN Mapping

Server	VLAN	Department	Router
S1	10	Sales	R1
S2	20	Finance	R2
S3	30	Manufacturing	R3
S4	99	Guest	R4

SW1 and SW2 are connected using a trunk link carrying VLANs 10, 20, 30 and 99

Figure 1: Final Network Topology



4. VLAN ADDRESSING PLAN

VLAN ID	Name	Network	Gateway	DHCP Range
10	Sales	192.168.10.0/24	192.168.10.1	192.168.10.50 – 192.168.10.200
20	Finance	192.168.20.0/24	192.168.20.1	192.168.20.50 – 192.168.20.150
30	Manufacturing	192.168.30.0/24	192.168.30.1	192.168.30.50 – 192.168.30.220
99	Guest Wi-Fi	192.168.99.0/24	192.168.99.1	192.168.99.50 – 192.168.99.250
999	Native	—	—	—

5. VLAN CONFIGURATION

Four VLANs were configured on both switches.

VLANs Configured

- VLAN 10 → SALES
- VLAN 20 → FINANCE
- VLAN 30 → MANUFACTURING
- VLAN 99 → GUEST
- VLAN 999 → NATIVE

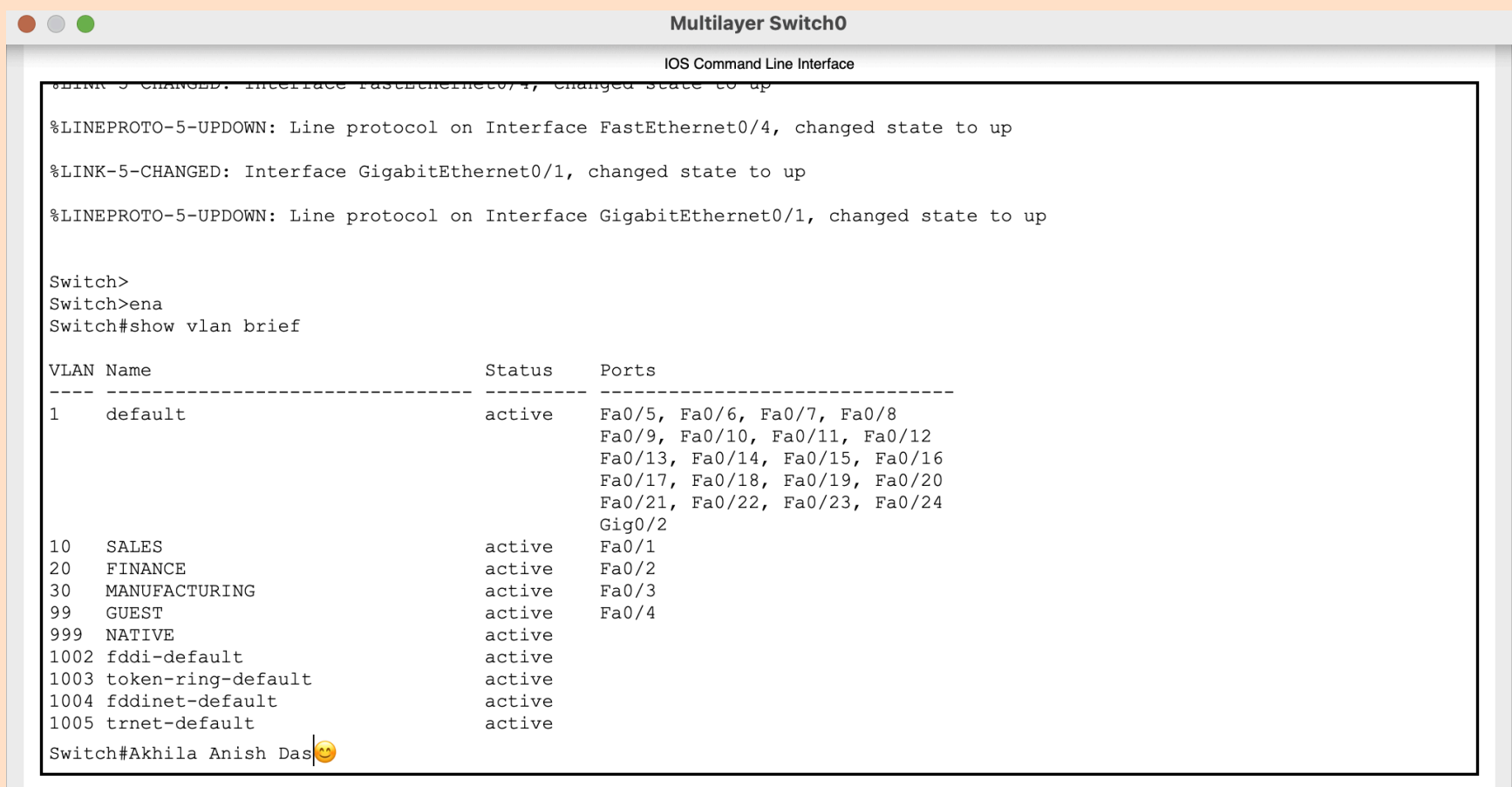
Access Port Assignment

- Fa0/1 → VLAN 10
- Fa0/2 → VLAN 20
- Fa0/3 → VLAN 30
- Fa0/4 → VLAN 99

SW1 Verification

Command used: ~ show vlan brief

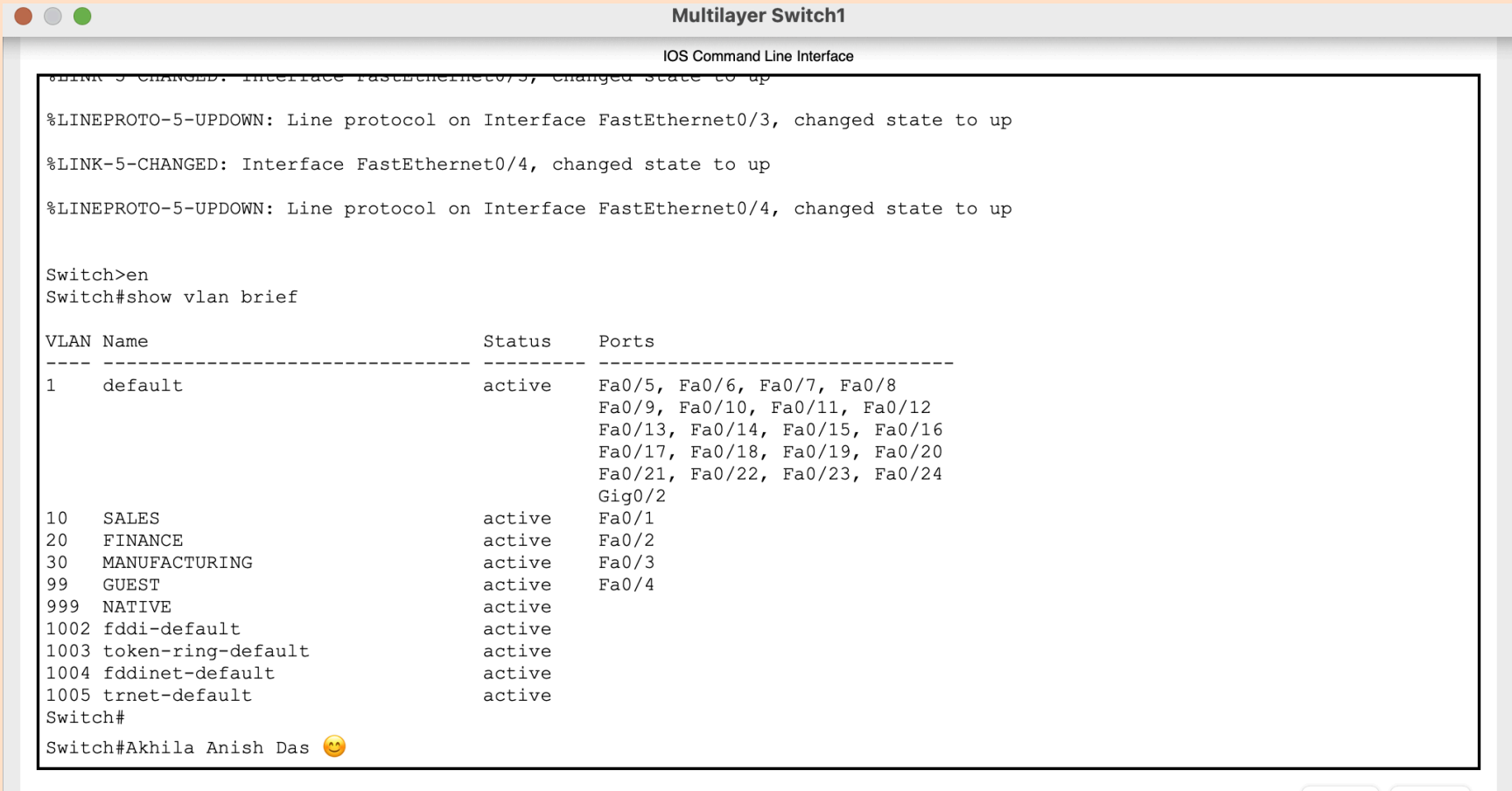
Figure 2: SW1 VLAN Configuration



SW2 Verification

Command used: ~ show vlan brief

Figure 3: SW2 VLAN Configuration



6. TRUNK CONFIGURATION

The connection between SW1 and SW2 was configured as an 802.1Q trunk.

The trunk carries the following VLANs:

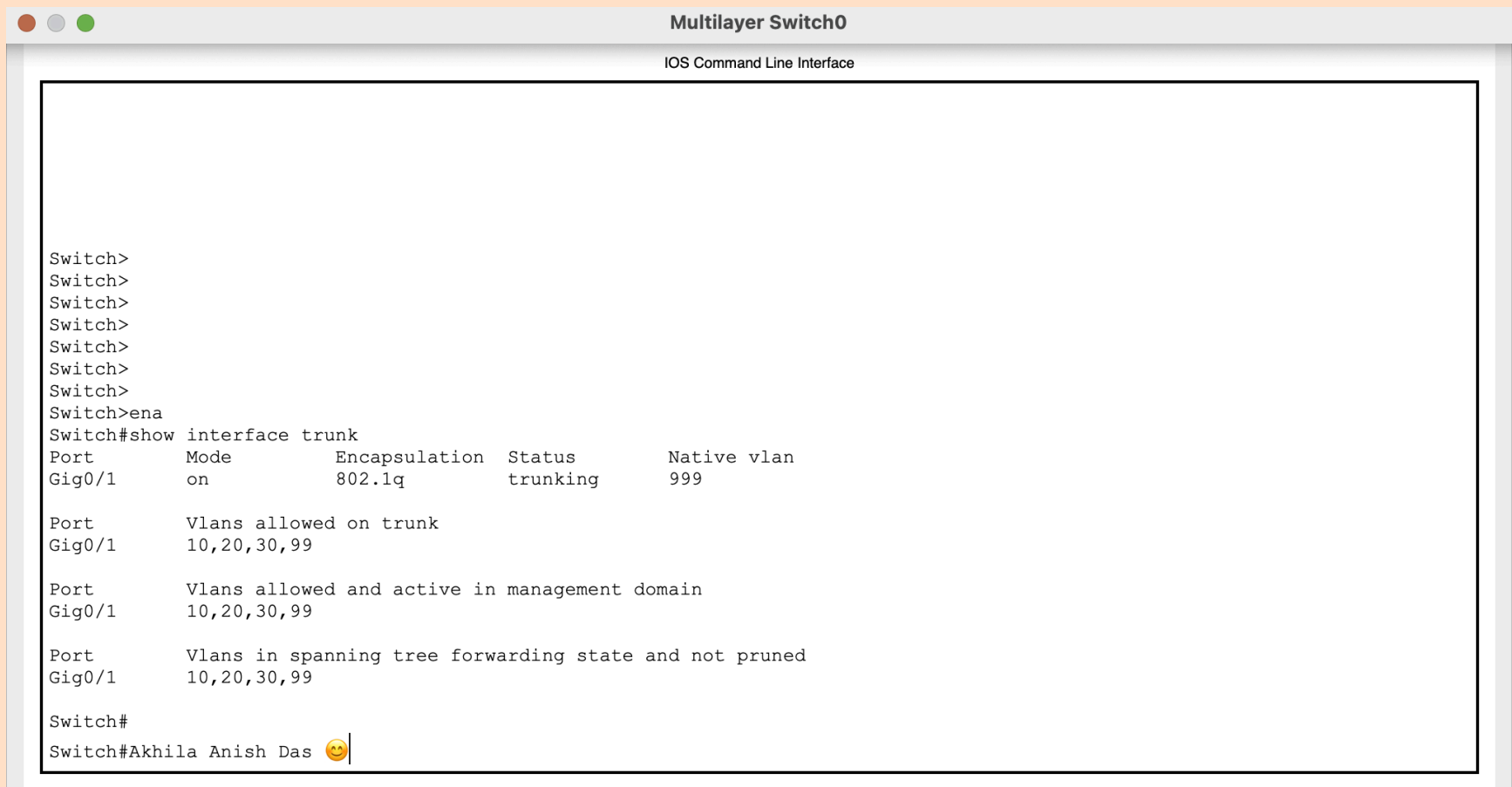
- VLAN 10
- VLAN 20
- VLAN 30
- VLAN 99

VLAN 999 was configured as the native VLAN.

SW1 Trunk Verification

Command used: ~ show interfaces trunk

Figure 4: SW1 Trunk Verification



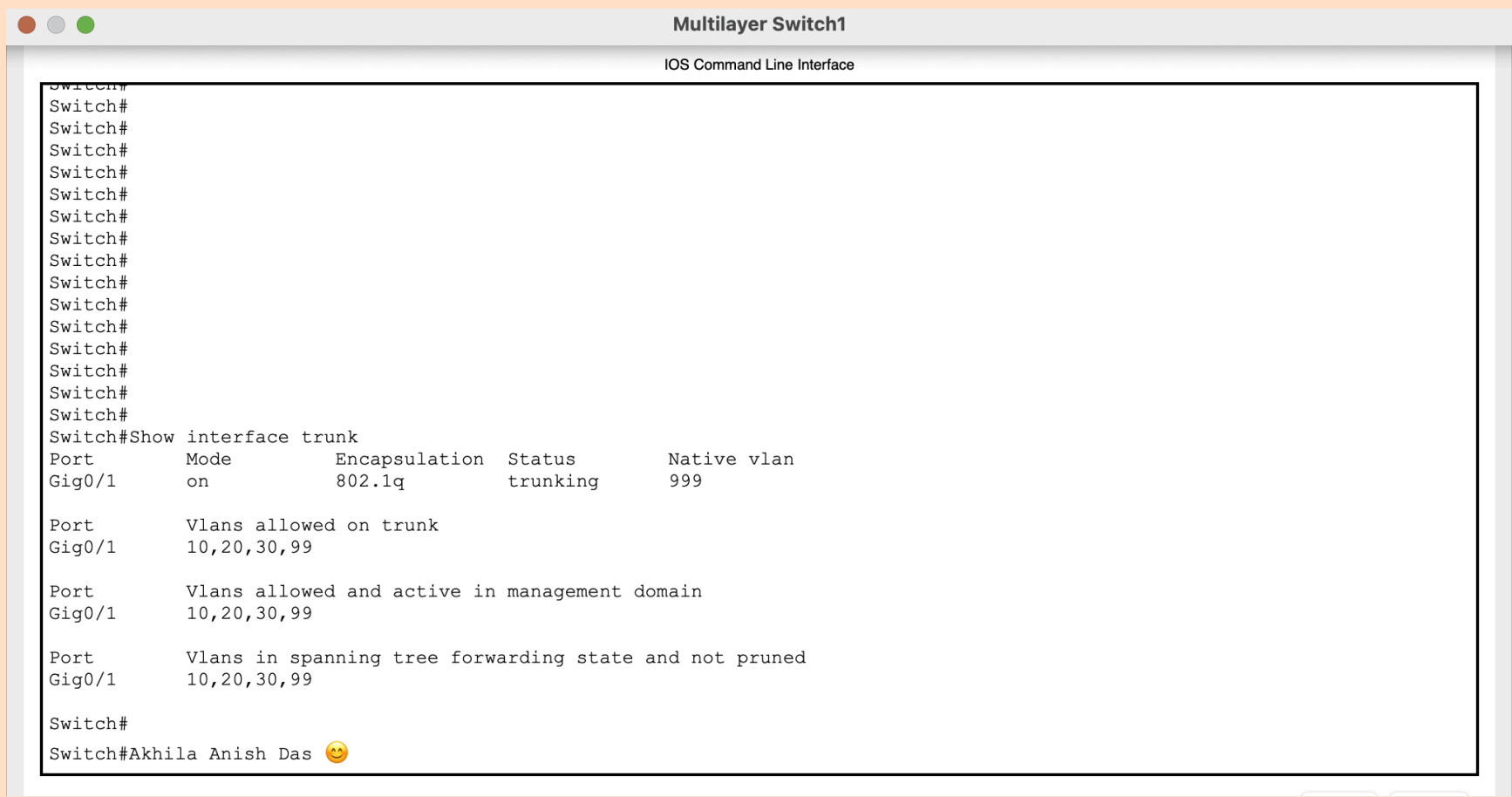
The output should show:

- Encapsulation: 802.1q
- Status: trunking
- Native VLAN: 999
- Allowed VLANs: 10,20,30,99

SW2 Trunk Verification

Command used: ~ show interfaces trunk

Figure 5: SW2 Trunk Verification



7. DHCP CONFIGURATION

DHCP was configured for each VLAN so that the servers could automatically receive IP addresses from their respective networks.

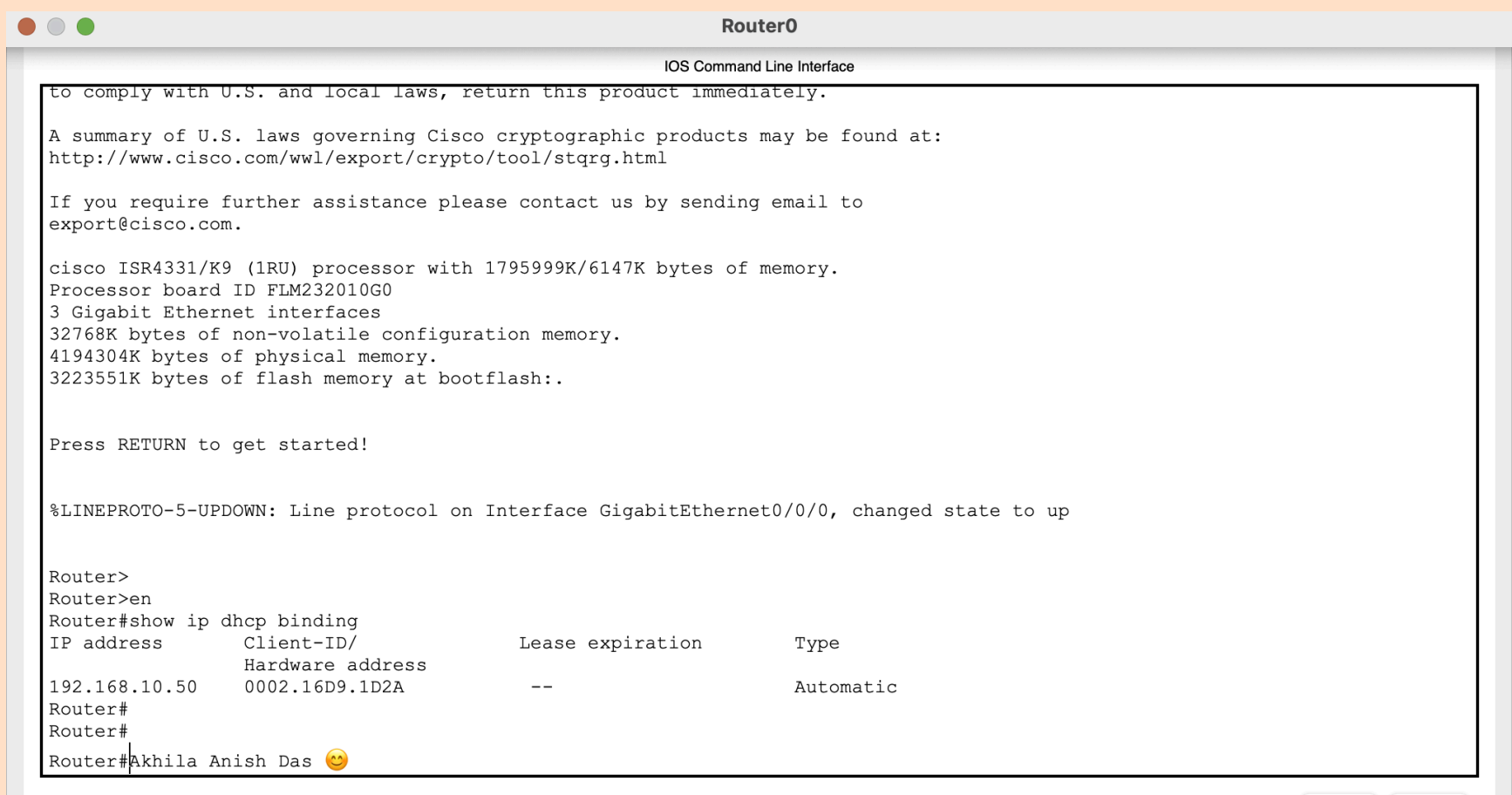
The DHCP networks are:

- VLAN 10 → 192.168.10.0/24
- VLAN 20 → 192.168.20.0/24
- VLAN 30 → 192.168.30.0/24

7.1 DHCP Verification – R1

Command used: ~ show ip dhcp binding

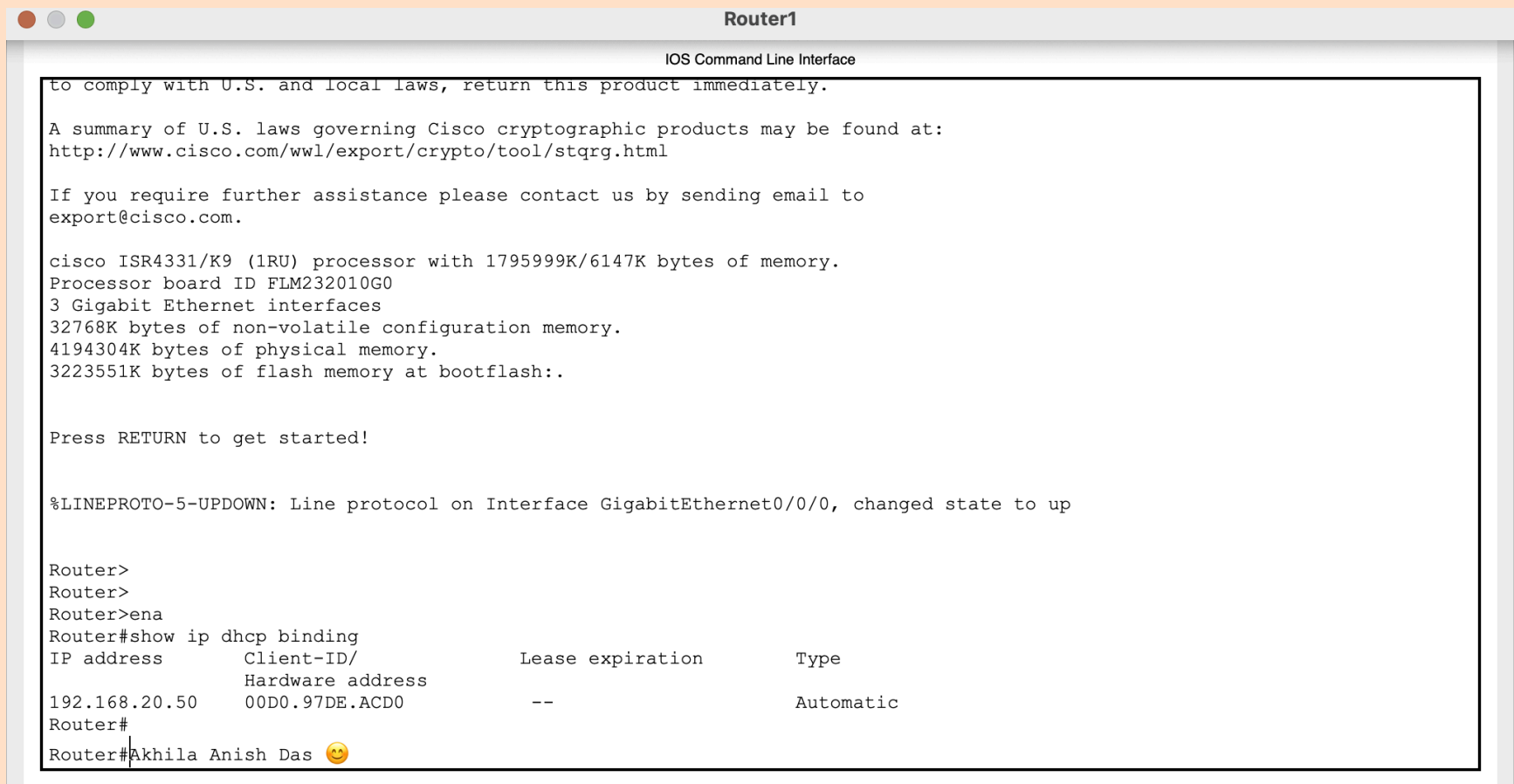
Figure 6: R1 DHCP Binding



7.2 DHCP Verification – R2

Command used: ~ show ip dhcp binding

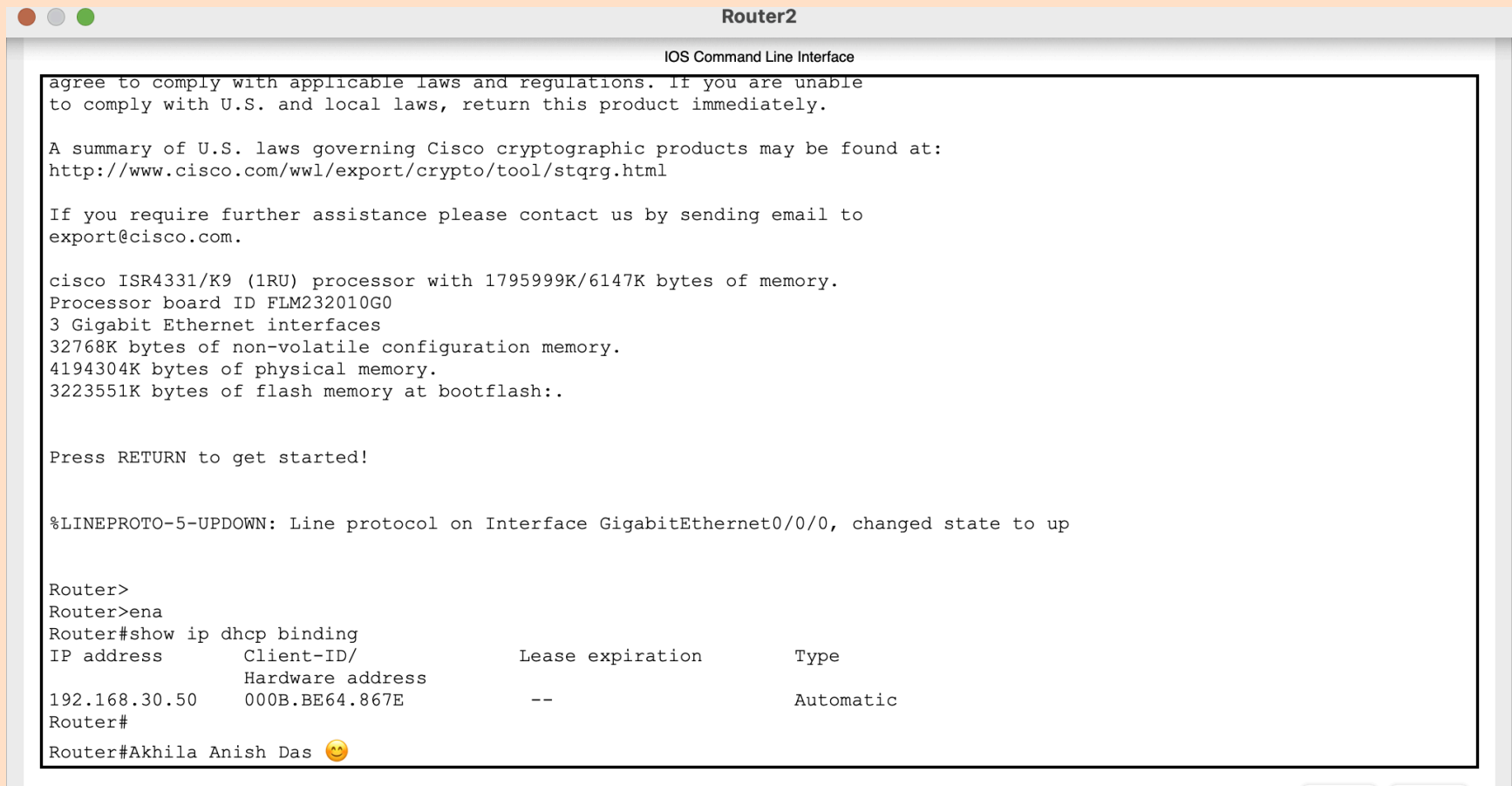
Figure 7: R2 DHCP Binding



7.3 DHCP Verification – R3

Command used: ~ show ip dhcp binding

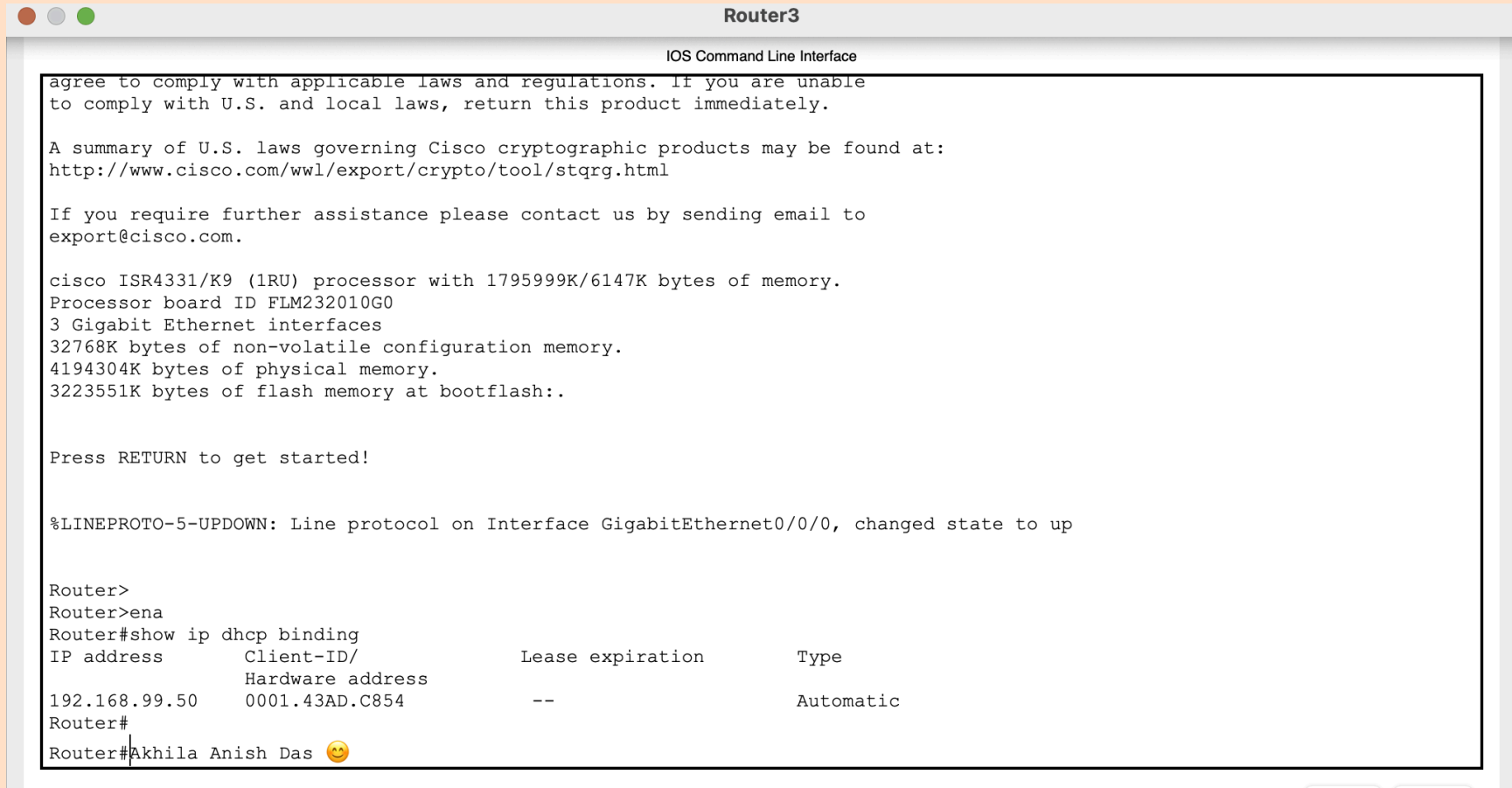
Figure 8: R3 DHCP Binding



7.4 DHCP Verification – R4

Command used: ~ show ip dhcp binding

Figure 9: R4 DHCP Binding



8. IP ADDRESS VERIFICATION

After DHCP configuration, each server was checked to verify that it received an IP address from the correct VLAN network.

Server	VLAN	Network
S1	10	192.168.10.x
S2	20	192.168.20.x
S3	30	192.168.30.x
S4	99	192.168.99.x

Example: ~ S1 → 192.168.10.50

9. CONNECTIVITY TESTING

Connectivity was tested between each server and its corresponding router.

The purpose of the test was to verify that each server could reach its VLAN gateway.

9.1 S1 to R1

Command: ~ ping 192.168.10.1

Figure 10: S1 to R1 Ping



Result: Successful communication with 0% packet loss.

9.2 S2 to R2

Command: ~ ping 192.168.20.1

Figure 11: S2 to R2 Ping

Server1

Command Prompt

X

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>Akhila Anish Das

Result: Successful communication with 0% packet loss.

9.3 S3 to R3

Command: ~ ping 192.168.30.1

Figure 12: S3 to R3 Ping

Server2

Command Prompt

X

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

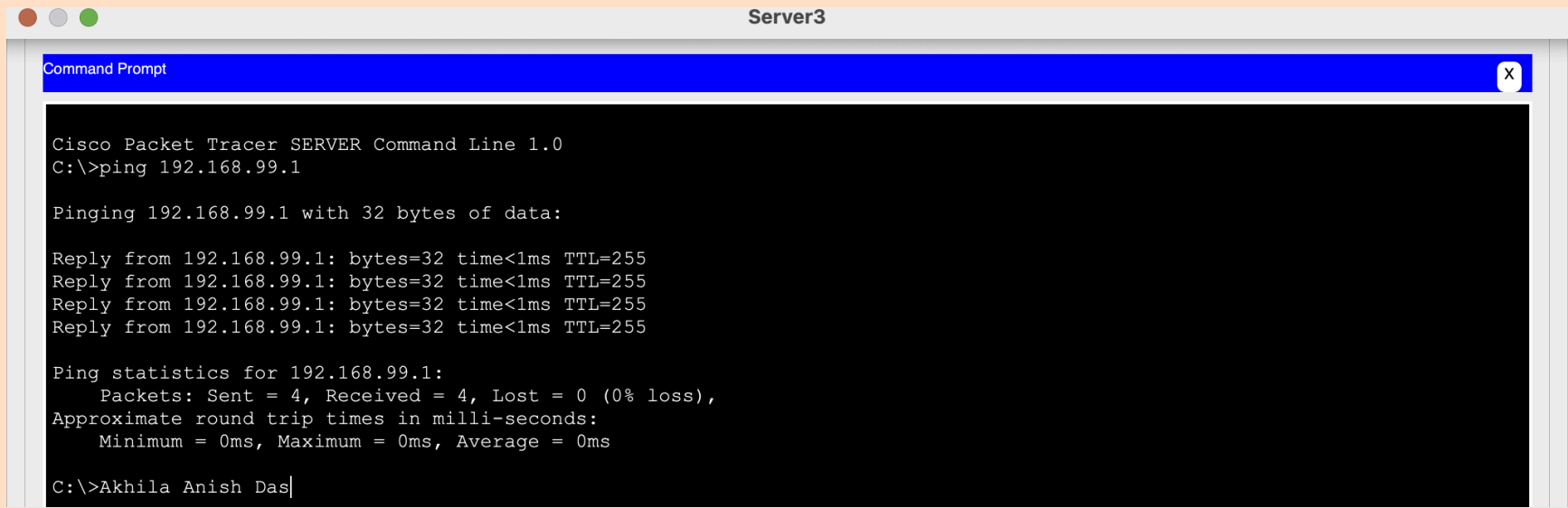
C:\>Akhila Anish Das|

Result: Successful communication with 0% packet loss.

9.4 S4 to R4

Command: ~ ping 192.168.99.1

Figure 13: S4 to R4 Ping



Result: Successful communication with 0% packet loss.

10. REVERSE CONNECTIVITY TESTING

Reverse connectivity was tested by pinging each server from its corresponding router.

R1 to S1

R1# ping 192.168.10.50

Expected result:

!!!!
Success rate is 100 percent (5/5)

Figure 14: R1 to S1 Ping

```
Router#Akhila Anish Das 😊
      ^
% Invalid input detected at '^' marker.

Router#
Router#ping 192.168.10.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.50, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

R2 to S2

R2# ping <S2-DHCP-IP>

Figure 15: R2 to S2 Ping

```
Router#Akhila Anish Das 😊
      ^
% Invalid input detected at '^' marker.

Router#
Router#ping 192.168.20.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.50, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

R3 to S3

R3# ping <S3-DHCP-IP>

Figure 16: R3 to S3 Ping

```
Router#Akhila Anish Das 😊
      ^
% Invalid input detected at '^' marker.

Router#
Router#ping 192.168.30.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.50, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

R4 to S4

R4# ping <S4-DHCP-IP>

Figure 17: R4 to S4 Ping

```
Router#Akhila Anish Das 😊
^
% Invalid input detected at '^' marker.

Router#
Router#ping 192.168.99.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.50, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

11. VERIFICATION SUMMARY

Test	Expected Result	Status
VLAN 10	VLAN active	PASS
VLAN 20	VLAN active	PASS
VLAN 30	VLAN active	PASS
VLAN 99	VLAN active	PASS
SW1–SW2 Trunk	Trunking	PASS
802.1Q Encapsulation	802.1Q	PASS
Native VLAN	999	PASS
VLAN 10 on Trunk	Allowed	PASS
VLAN 20 on Trunk	Allowed	PASS
VLAN 30 on Trunk	Allowed	PASS

VLAN 99 on Trunk	Allowed	PASS
DHCP VLAN 10	Address Assigned	PASS
DHCP VLAN 20	Address Assigned	PASS
DHCP VLAN 30	Address Assigned	PASS
DHCP VLAN 99	Address Assigned	PASS
S1 → R1	Successful	PASS
S2 → R2	Successful	PASS
S3 → R3	Successful	PASS
S4 → R4	Successful	PASS

12. EXPECTED OUTCOME

The completed network demonstrates:

1. Separate VLANs for each department.
2. Automatic IP address assignment through DHCP.
3. Successful communication between each server and its corresponding gateway.
4. A working 802.1Q trunk between SW1 and SW2.
5. VLANs 10, 20, 30 and 99 carried across the trunk.
6. Native VLAN 999 configured on the trunk.
7. DHCP-based IP assignment without requiring manual IP configuration.

13. CONCLUSION

The network was successfully configured using VLANs, DHCP and an 802.1Q trunk link in Cisco Packet Tracer.

Four separate VLANs were created for Sales, Finance, Manufacturing and Guest Wi-Fi. DHCP was configured for automatic IP address assignment, while the SW1–SW2 connection was configured as a trunk carrying VLANs 10, 20, 30 and 99 with VLAN 999 as the native VLAN.

Connectivity testing confirmed that each server could communicate with its corresponding router gateway.

14. DISCUSSION QUESTIONS

1. Why does the link between the core switch and the warehouse switch need to be a trunk instead of an access port?

The link between the core switch and the warehouse switch needs to be a **trunk link** because it carries traffic from all four VLANs simultaneously. A trunk allows multiple VLANs to pass through a single physical connection, whereas an access port is associated with only one VLAN. This avoids the need for separate connections for each VLAN.

2. What would happen to Guest Wi-Fi traffic if the trunk's allowed VLAN list did not include VLAN 99?

If **VLAN 99** is not included in the trunk's allowed VLAN list, Guest Wi-Fi traffic cannot pass through the trunk. As a result, Guest VLAN devices connected through the other switch would not be able to communicate across that trunk. The Guest VLAN would therefore not function correctly across the two switches.

3. Why is it useful for each VLAN to have its own DHCP scope rather than one shared scope for the whole company?

Each VLAN represents a separate network and therefore requires IP addresses from its own subnet. A separate DHCP scope ensures that devices in each VLAN receive the correct IP address range and gateway.

For example:

VLAN 10 → 192.168.10.x

VLAN 20 → 192.168.20.x

VLAN 30 → 192.168.30.x

VLAN 99 → 192.168.99.x

This keeps IP addressing organized and ensures that devices receive addresses appropriate to their respective VLAN.

4. What risk does changing the native VLAN to an unused ID (999) help reduce?

Changing the native VLAN to an **unused VLAN ID such as 999** helps reduce the risk of unwanted traffic using the default/native VLAN. Using an unused VLAN for native traffic provides better separation from normal user VLANs and reduces the possibility of accidental or unauthorized traffic being carried on the native VLAN.

15. SUBMISSION FILES

VLAN-DHCP-TRUNK-ASSIGNMENT

- Documentation.pdf
- Verification-Screenshots
- Network-Topology.pkt

16. AUTHOR

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